

**12.4 Classwork: de Moivre, Trigonometric Identities and Roots of Unity —
Markscheme**

1. [Maximum mark: 6]

(a) $(\cos \theta + i \sin \theta)^3 = \cos 3\theta + i \sin 3\theta$ **A1**

A1 for a correct statement of de Moivre's theorem at $n = 3$. Accept $\text{cis } 3\theta$ or $e^{3i\theta}$ written alongside.

(b) $(\cos \theta + i \sin \theta)^3 = \cos^3 \theta + 3 \cos^2 \theta (i \sin \theta) + 3 \cos \theta (i \sin \theta)^2 + (i \sin \theta)^3$ **M1**

$= \cos^3 \theta + 3i \cos^2 \theta \sin \theta - 3 \cos \theta \sin^2 \theta - i \sin^3 \theta$ **A1**

M1 for a complete binomial expansion with the four terms and the correct coefficients 1, 3, 3, 1, however it is set out. A1 for simplifying the powers of i correctly ($i^2 = -1$, $i^3 = -i$) and collecting into real and imaginary parts. An expansion that keeps i^2 unevaluated earns the M1 only.

(c) Equating real parts with (a): $\cos 3\theta = \cos^3 \theta - 3 \cos \theta \sin^2 \theta$ **M1**

$= \cos^3 \theta - 3 \cos \theta (1 - \cos^2 \theta)$ **A1**

$= \cos^3 \theta - 3 \cos \theta + 3 \cos^3 \theta = 4 \cos^3 \theta - 3 \cos \theta$ **AG**

M1 for equating the real part of (a) with the real part of (b); A1 for using $\sin^2 \theta = 1 - \cos^2 \theta$ to remove the sine. The AG is the printed result, so the two lines before it must both be present — a candidate who writes the answer straight down from memory earns nothing here.

2. [Maximum mark: 5]

(a) Equating imaginary parts: $\sin 3\theta = 3\cos^2 \theta \sin \theta - \sin^3 \theta$ M1

$= 3(1 - \sin^2 \theta) \sin \theta - \sin^3 \theta = 3\sin \theta - 4\sin^3 \theta$ A1

M1 for equating imaginary parts of the same two expressions; A1 for the expression in $\sin \theta$ only. Stopping at $3\cos^2 \theta \sin \theta - \sin^3 \theta$ earns the M1 only, since the question asks for $\sin \theta$ alone.

(b) $8\cos^3 \theta - 6\cos \theta = 2(4\cos^3 \theta - 3\cos \theta) = 2\cos 3\theta$, so $\cos 3\theta = \frac{1}{2}$ M1

$0 \leq \theta < \pi \Rightarrow 0 \leq 3\theta < 3\pi$, and $\cos x = \frac{1}{2}$ at $x = \frac{\pi}{3}, \frac{5\pi}{3}, \frac{7\pi}{3}$ A1

$\theta = \frac{\pi}{9}, \frac{5\pi}{9}, \frac{7\pi}{9}$ A1

*M1 for recognising the left side as $2\cos 3\theta$ using the identity from Problem 1 — this is the whole point of the “hence”. A1 for finding all three values of 3θ in the widened interval $[0, 3\pi)$; a candidate who finds only $3\theta = \pi/3$ loses this mark. A1 for the three values of θ . **FT:** award the final A1 for the candidate’s own 3θ values each correctly divided by 3. A candidate who solves the cubic numerically earns nothing: the question says “hence”, and exact multiples of π are required.*

3. [Maximum mark: 5]

(a) $z^5 = 1 = e^{2k\pi i}$, so $z = e^{2k\pi i/5}$, $k \in \mathbb{Z}$ **M1**

Taking $k = 0, \pm 1, \pm 2$ to place every argument in $-\pi < \theta \leq \pi$: **A1**

$$z = 1, \quad e^{2\pi i/5}, \quad e^{4\pi i/5}, \quad e^{-2\pi i/5}, \quad e^{-4\pi i/5}$$

A1

M1 for writing 1 in exponential form with the $2k\pi$ allowance and taking fifth roots. First A1 for the modulus 1 together with any one correct non-zero argument. Second A1 for all five roots with every argument inside the stated range — $e^{6\pi i/5}$ and $e^{8\pi i/5}$ are outside it and must be converted to $e^{-4\pi i/5}$ and $e^{-2\pi i/5}$. Accept cis form throughout.

(b) The five roots are the roots of $z^5 - 1 = 0$, in which the coefficient of z^4 is 0, so their sum is 0. **R1**

Accept any complete argument: the sum of roots from the coefficients, as above; the geometric series $1 + \omega + \omega^2 + \omega^3 + \omega^4 = \frac{1 - \omega^5}{1 - \omega} = \frac{1 - 1}{1 - \omega} = 0$ for $\omega \neq 1$; or the symmetry argument that five equal vectors spaced $2\pi/5$ apart cancel. Asserting “they cancel by symmetry” with nothing further earns nothing — the equal spacing has to be the stated reason.

(c) A **regular pentagon**, centred at the origin, with every vertex a distance 1 from the origin (the roots lie on the unit circle, one vertex at $z = 1$, consecutive vertices $\frac{2\pi}{5}$ apart). **A1**

A1 requires the pentagon and both requested details — centre at the origin and vertices at distance 1. “A pentagon” alone earns nothing, and neither does “a regular polygon” without the number of sides.